

Week_7

```
srikan@DESKTOP-0NIMS2G: ~/OSSP
srikan@DESKTOP-0NIMS2G:~/OSSP$ gcc client.c -o client
srikan@DESKTOP-0NIMS2G:~/OSSP$ ./server
Server started. Waiting for client...
Client: hello
Client: hello
^Z
[3]+  Stopped                  ./server
srikan@DESKTOP-0NIMS2G:~/OSSP$ ./client
Enter message: hello
Server: Server received: hello
Enter message: hihihihihihihihhi
Server: Server received: hihihihihihihihhi
Enter message: ^Z
[4]+  Stopped                  ./client
srikan@DESKTOP-0NIMS2G:~/OSSP$ gedit week_7.c
srikan@DESKTOP-0NIMS2G:~/OSSP$ gcc week_7.c -o week_7
srikan@DESKTOP-0NIMS2G:~/OSSP$ ./week_7
===== Process Memory Addresses =====
Code address   : 0x6303bec08209
Global address  : 0x6303bec0b010
Static address  : 0x6303bec0b014
Heap address    : 0x6303d1ab02a0
Stack address   : 0x7f7ea23892fc

Process ID (PID): 743

Process is running...
Use another terminal to check /proc/743/maps
Press Enter to exit...

srikan@DESKTOP-0NIMS2G:~/OSSP$
srikan@DESKTOP-0NIMS2G:~/OSSP$ gedit week_7.c
```

```
Open  week_7.c
Save  X

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4
5 int global_var = 10;           // Global variable
6 static int static_var = 20;    // Static variable
7
8 void code_function() {
9     printf("This function is in the code segment.\n");
10 }
11
12 int main() {
13     int stack_var = 30;         // Stack variable
14     int *heap_var;              // Pointer
15
16     heap_var = (int *)malloc(sizeof(int));
17     *heap_var = 40;
18
19     printf("===== Process Memory Addresses =====\n");
20
21     printf("Code address   : %p\n", (void *)code_function());
22     printf("Global address  : %p\n", (void *)&global_var);
23     printf("Static address  : %p\n", (void *)&static_var);
24     printf("Heap address    : %p\n", (void *)heap_var);
25     printf("Stack address   : %p\n", (void *)&stack_var);
26
27     printf("\nProcess ID (PID): %d\n", getpid());
28
29     printf("\nProcess is running...\n");
30     printf("Use another terminal to check /proc/%d/maps\n", getpid());
31
32     printf("Press Enter to exit...\n");
33     getchar();
34
35     free(heap_var);
36
37     return 0;
38 }
```

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